

Chapter # 2

Chapter 2

CLO-2

DIFFERENTIAL CALCULUS

Differentiation

$$\frac{dy}{dx}, \frac{d}{dx} f(x)$$

Successive
Diff.

$$\frac{d^2}{dx^2}, \frac{d^3}{dx^3}$$

BASIC TECHNIQUES:-

$$\rightarrow \frac{d}{dx} [c] = 0 \quad \text{constant} \quad \rightarrow \frac{d}{dx} [f(x) + g(x)] = \frac{d}{dx} f(x) + \frac{d}{dx} g(x)$$

$$\rightarrow \frac{d}{dx} [x] = 1$$

$$\rightarrow \frac{d}{dx} [x^n] = nx^{n-1}$$

$$\rightarrow \frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{g(x) \frac{d}{dx} f(x) - f(x) \frac{d}{dx} g(x)}{[g(x)]^2}$$

$$\rightarrow \frac{d}{dx} [f(x) \cdot g(x)] = f(x) \frac{d}{dx} g(x) + g(x) \frac{d}{dx} f(x)$$

$$Q_1: \lim_{x \rightarrow 0} \frac{e^x - 1}{x^3} \begin{matrix} \nearrow f'(x) \\ \longrightarrow g'(x) \end{matrix}$$

$$\frac{e^x}{3x^2} \Rightarrow \frac{1}{0} \Rightarrow \boxed{\infty} \text{ Ans.}$$

$$Q_2: \lim_{x \rightarrow \infty} \frac{x^{-4/3}}{\sin(1/x)} \rightarrow \frac{d}{dx} \frac{x^{-4/3}}{\sin(1/x)}$$

$$= \frac{-4/3 x^{-7/3}}{-x^{-2} \cos(1/x)}$$

$$= \frac{x^{-4/3}}{\cos(1/x)}$$

$$\frac{\infty^{-4/3}}{\cos(1/\infty)}$$

$$\frac{0}{1} \Rightarrow \boxed{0} \text{ Ans.}$$

$$Q3: \lim_{x \rightarrow 0} \frac{e^x - e^{-x}}{\sin x}$$

$$\frac{d}{dx} \frac{e^x - e^{-x}}{\sin x}$$

$$\Rightarrow \frac{e^x + e^{-x}}{\cos x}$$

$$\Rightarrow \frac{e^0 + e^0}{\cos 0}$$

$$\Rightarrow \frac{1+1}{1}$$

$$\boxed{2} \text{ Ans}$$

$$Q4: \lim_{x \rightarrow \infty} \frac{e^{x^2} - 1}{\cos x - 1}$$

$$\frac{d}{dx} \frac{e^{x^2} - 1}{\cos x - 1}$$

$$\Rightarrow \frac{d}{dx} \frac{2xe^{x^2}}{-\sin x}$$

$$\Rightarrow \frac{2[x \cdot 2xe^{x^2} + e^{x^2} \cdot 1]}{-\cos x}$$

$$\Rightarrow \frac{2[1]}{-\cos 0}$$

$$\Rightarrow \boxed{-2} \text{ Ans}$$

wed, Jan 28

$$\lim_{x \rightarrow 0} \frac{x - \tan(x)}{x - \sin x} = \frac{0}{0}$$

$$d(x) = \frac{1 - \sec^2 x}{1 - \cos x} = \frac{0}{0}$$

$$\begin{aligned} d^2(x) &= \frac{0 - 2 \sec x (\sec x \cdot \tan x)}{0 - (-\sin x)} \\ &= \frac{-2 \sec x (\sec x \cdot \tan x)}{\sin x} = \frac{0}{0} \end{aligned}$$

$$\tan = \frac{\sin x}{\cos x}$$

$$\sec = \frac{\cos x}{\sin x}$$

$$= -2 \cdot \frac{1}{\cos x} \cdot \frac{1}{\cos x} \cdot \frac{\sin x}{\cos x} \quad \checkmark \quad \frac{1}{\cancel{\cos x}}$$

$$= \frac{-2}{\cos^3 x}$$

$$= -2 \quad \text{Ans}$$

$$\text{Q2} \quad \lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^2 \tan x} = \frac{0}{0}$$

$$v = x^2 \quad u = \tan x$$

$$= \frac{\sec^2 x - \cos x}{2x \tan x + x^2 \sec^2 x} = \frac{0}{0}$$

$$\begin{aligned} &uv' + vu' \\ &= \tan x(2x) + x^2 \cdot \sec^2 x \end{aligned}$$

$$\begin{aligned} \frac{\tan x - \sin x}{x^2 \tan x} &= \frac{\frac{\sin x}{\cos x} - \sin x}{\frac{\sin x - \sin x \cos x}{\cos x}} \div x^2 \times \frac{\sin x}{\cos x} \\ &= \frac{\sin x - \sin x \cos x}{\cos x} \div x^2 \times \frac{\sin x}{\cos x} \end{aligned}$$

$$\begin{aligned}
 & \frac{\sin x - \sin x \cos x}{\cos x} \times \frac{\cancel{\cos x}}{x^2 \sin x} \\
 &= \frac{\sin x - \sin x \cos x}{x^2 \sin x} \\
 &= \frac{\cancel{\sin x} (1 - \cos x)}{x^2 \cancel{\sin x}} \\
 &= \frac{1 - \cos x}{x^2} = \frac{0}{0} \\
 &= \frac{0 - (-\sin x)}{2x} \\
 &= \frac{\sin x}{2x} \\
 &= \frac{\cos x}{2} = \frac{1}{2} \text{ Ans}
 \end{aligned}$$

Q) $\lim_{x \rightarrow 0} \frac{\ln(\sin 3x)}{\ln(\sin x)} =$

$$\begin{aligned}
 & \frac{1}{\sin 3x \cdot 3 \cos 3x} \times \sin x \cos x \\
 & \sin x \cdot 3 \cos 3x \\
 & 3 \sin x \cdot \sin 3x
 \end{aligned}$$

$$\left| \begin{aligned}
 & \ln \left(\frac{\sin 3x}{\sin x} \right) \\
 & \ln \left(\frac{3 \cos 3x}{\cos x} \right)
 \end{aligned} \right.$$

$\cos(0) = 1$
 $\ln(3) = 1.1$

